Can You Hear the Shape of an Organ? Practical Identifiability of Viscoelastic Shell Parameters from Resonant Frequencies

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Abstract

We investigate the practical identifiability of material and geometric parameters of a fluid-filled viscoelastic oblate spheroidal shell from its resonant frequencies — the inverse counterpart of Kac's celebrated question "Can one hear the shape of a drum?" applied to vibroacoustic organ models. Using an equivalent-sphere flexural forward model, we compute the scaled Jacobian of the frequency–parameter map, its singular values, its condition number, and the associated Fisher information matrix for the parameter triple (a, c, E). At the spherical operating point the scaled Jacobian is effectively rank deficient (condition number $\kappa = 1.37 \times 10^{10}$); at the canonical oblate operating point the same analysis gives $\kappa = 69.4$, an improvement by a factor of 2.0×10^8. A Newton-type inverse solver recovers (a, c, E) to relative errors below 10^{-12} in round-trip tests, while Cramér–Rao bounds at 1% frequency noise yield uncertainties of ±22.5% in a, ±6.0% in c, and ±30.8% in E. Five-mode inversion is robust; three-mode inversion is vulnerable to local minima. The result has implications for non-invasive tissue characterisation and fruit quality assessment: one can "hear the shape" of an oblate organ, but not a spherical one.

Keywords: inverse problem, identifiability, viscoelastic shell, resonant frequency, condition number, Cramér–Rao bound, oblate spheroid, Kac problem

1. Introduction

In 1966, Kac asked the now-classic question "Can one hear the shape of a drum?" [1]. The question is spectral: does the set of resonant frequencies determine the geometry uniquely? Gordon, Webb, and Wolpert [2] later showed that, for planar drums, the answer is "no": distinct domains can share the same Laplacian spectrum. That result did not close the subject. It made it more interesting.

The Browntone programme has approached that question from the forward side. Paper 1 established that the human abdomen supports low-order flexural resonances in the infrasonic band [3]; Paper 4 extended the same fluid–shell framework to the fill-dependent resonance of the urinary bladder [4]; and Paper 7 showed that an analogous model can invert a watermelon tap-tone into a rind modulus estimate [5]. Those papers answer variants of the question "what frequencies does this shell produce?" The present paper asks the inverse question on which all such applications ultimately depend: given measured frequencies, can one recover the shell's material properties at all?

That identifiability question is the theoretical keystone of the programme. Without it, a forward resonance model is descriptive but not diagnostic. With it, the same spectrum can become the basis of non-invasive elastography,

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parameter estimation, and experimental design. In practical terms, if a small set of resonant frequencies determines stiffness and geometry with acceptable conditioning, then the abdominal, bladder, and watermelon problems cease to be curiosities of shell vibration and become inverse tools. If not, one merely fits a model to a spectrum that never had enough information in the first place.

Here we therefore study an engineering analogue of Kac's inverse problem for a fluid-filled viscoelastic shell. We focus on low-frequency flexural modes $n \geq 2$, rather than the breathing mode $n = 0$, which lies near 2490 Hz and is governed primarily by fluid compressibility. The inverse problem is posed on the measured flexural frequencies $\{f_2, f_3, \ldots, f_N\}$, with the aim of recovering the semi-major axis a, the semi-minor axis c, and the Young's modulus E.

The key geometric point is oblateness. A spherical shell sits at a spectrally degenerate limit: symmetry forces geometric sensitivities to collapse, so the inverse map becomes singular or nearly so. An oblate shell does not enjoy that luxury. Once $a \neq c$, different flexural modes sample curvature differently, the degeneracy is lifted, and identifiability can in principle be restored. The question is not whether asymmetry looks promising in a qualitative sense, but whether it restores enough rank to support useful inversion.

A central modelling point must also be stated plainly. The implemented forward model is an *equivalent-sphere* model. It uses the equivalent radius

$$R_{\mathrm{eq}} = (a^2 c)^{1/3}, \tag{1}$$

and does not contain explicit spheroidal curvature terms or explicit aspect-ratio corrections. We therefore do *not* claim a complete oblate shell theory in the numerical sections. Rather, we use the equivalent-sphere model to expose the inverse structure already visible in the Browntone applications, and we pair it with analytical results that show why a genuine oblate Ritz model should restore identifiability away from the spherical limit.

Our contribution is therefore twofold. First, we quantify practical identifiability through the Jacobian of the frequency–parameter map, using the dimensionless scaled Jacobian, its singular values, its condition number, and the associated Fisher information matrix to assess whether small perturbations in the measured frequencies can be mapped reliably back to (a, c, E). Second, we make explicit the programme-level consequence: identifiability, not merely forward prediction, is the mathematical foundation on which Papers 1–7 rest. This is an analysis of *practical* identifiability in the sense of Raue et al. [6] and Wieland et al. [7], not a global uniqueness theorem.

The main numerical findings are as follows.

- At the spherical operating point, the scaled Jacobian is effectively rank deficient, with condition number $\kappa = 1.37 \times 10^{10}$.
- At the canonical oblate operating point, the same analysis gives $\kappa = 69.4$, an improvement by a factor of 2.0×10^8.
- A Newton-type inverse solver recovers (a, c, E) to relative errors below 10^{-12} in round-trip tests when initialised within the local basin of attraction.
- At 1% frequency noise, the Cramér–Rao lower bounds correspond to uncertainties of approximately ±22.5% in a, ±6.0% in c, and ±30.8% in E.

- Three-mode inversion is locally feasible but exhibits local minima, whereas five-mode inversion is robust across the tested initial guesses.

These results suggest that one can, in a practical sense, hear the shape of an oblate organ from a modest set of flexural resonances, whereas the same task is poorly posed at the spherical limit. For the Browntone programme, that distinction is not a side remark. It explains why the forward models of Papers 1–7 are potentially invertible in the first place, and why oblateness rather than near-sphericity is the feature that turns resonance from a descriptive quantity into a usable diagnostic observable.

The paper is organised as follows. Section 2 summarises the equivalent-sphere forward model and the inverse formulation. Section 3 describes the numerical Jacobian, the Newton-type inversion procedure, and the parameter-space conditioning study. Section 4 presents the sphere–oblate contrast, singular-value structure, Cramér–Rao bounds, and a watermelon cross-application. Section 5 relates the results back to Kac's question, the Browntone applications, and vibroacoustic elastography, and closes with three formal conjectures that emerge from the numerics. Section 6 summarises the practical recommendations and the programme-level implications.

2. Theory

2.1. Equivalent-sphere forward model

We use the flexural-frequency model introduced in the companion paper [3], drawing on classical shell vibration theory [8, 9]. The shell has semi-major axis a, semi-minor axis c, thickness h, Young's modulus E, Poisson's ratio ν, wall density ρ_w, and internal fluid density ρ_f. The fluid bulk modulus is denoted by K_f, the static internal pressure by P_{iap}, and the loss tangent by η. Unless stated otherwise, the canonical parameter set is

$$a = 0.18\,\mathrm{m}, \quad c = 0.12\,\mathrm{m}, \quad h = 0.01\,\mathrm{m}, \quad E = 0.1\,\mathrm{MPa}, \quad \nu = 0.45,$$

$$\rho_w = 1100\,\mathrm{kg/m^3}, \quad \rho_f = 1020\,\mathrm{kg/m^3}, \quad K_f = 2.2\,\mathrm{GPa}, \quad P_{\mathrm{iap}} = 1000\,\mathrm{Pa}, \quad \eta = 0.25.$$

The model collapses the geometry to the equivalent radius

$$R_{\mathrm{eq}} = (a^2 c)^{1/3}. \tag{2}$$

For flexural modes $n \geq 2$, the angular frequency $\omega_n = 2\pi f_n$ is written as

$$\omega_n^2 = \frac{K_{\mathrm{bend}}^{(n)} + K_{\mathrm{memb}}^{(n)} + K_{\mathrm{pre}}^{(n)}}{m_{\mathrm{eff}}^{(n)}}, \tag{3}$$

where the effective stiffness contains bending, membrane, and prestress terms, and the effective mass contains wall mass and fluid added mass.

With flexural rigidity

$$D = \frac{Eh^3}{12(1-\nu^2)}, \tag{4}$$

the individual contributions are

$$K_{\text{bend}}^{(n)} = \frac{n(n-1)(n+2)^2}{R_{\text{eq}}^4} D, \tag{5}$$

$$K_{\text{memb}}^{(n)} = \frac{Eh}{R_{\text{eq}}^2} \lambda_n, \qquad \lambda_n = \frac{n^2+n-2+2\nu}{n^2+n+1-\nu}, \tag{6}$$

$$K_{\text{pre}}^{(n)} = \frac{P_{\text{iap}}}{R_{\text{eq}}}(n-1)(n+2), \tag{7}$$

and

$$m_{\text{eff}}^{(n)} = \rho_w h + \frac{\rho_f R_{\text{eq}}}{n}. \tag{8}$$

Equation (3) is the model analysed throughout this paper. It is important to emphasise again that it is *not* a full oblate spheroidal shell theory: explicit aspect-ratio terms do not appear. The model depends on (a, c) only through R_{eq}.

2.2. Inverse problem

Let

$$\boldsymbol{\theta} = (a, c, E)^\top \tag{9}$$

denote the parameter vector to be estimated, and let

$$\mathbf{f}(\boldsymbol{\theta}) = \big(f_2(\boldsymbol{\theta}), f_3(\boldsymbol{\theta}), \ldots, f_N(\boldsymbol{\theta})\big)^\top \tag{10}$$

be the vector of measured flexural frequencies. The inverse problem is to recover $\boldsymbol{\theta}$ from the observation $\mathbf{f}^{\text{obs}}$.

Local identifiability is controlled by the Jacobian

$$J_{ij} = \frac{\partial f_i}{\partial \theta_j}, \tag{11}$$

where i indexes the measured mode and j indexes the unknown parameter. Because the raw Jacobian mixes units of Hz/m and Hz/Pa, its condition number is not a meaningful identifiability metric by itself. We therefore use the dimensionless scaled Jacobian

$$J_s[i, j] = \left(\frac{\partial f_i}{\partial \theta_j}\right)\left(\frac{\theta_j}{f_i}\right), \tag{12}$$

which measures fractional frequency change per fractional parameter change.

The singular values $\sigma_1 \geq \sigma_2 \geq \sigma_3$ of $\mathbf{J}_s$ quantify the observable directions in parameter space, and the corresponding condition number

$$\kappa(\mathbf{J}_s) = \frac{\sigma_{\max}}{\sigma_{\min}} \tag{13}$$

measures the worst-case amplification of small frequency errors into parameter errors. A large κ indicates a nearly unobservable parameter combination; a divergent κ indicates rank deficiency.

2.3. Fisher information and Cramér–Rao bounds

Assume independent fractional frequency errors with standard deviation ϵ_{noise}. In dimensionless form, the Fisher information matrix becomes

$$\mathbf{F} = \frac{1}{\epsilon_{\text{noise}}^2} \mathbf{J}_s^\top \mathbf{J}_s. \tag{14}$$

Equivalently, one may interpret (14) as the Fisher matrix of the whitened residuals. The Cramér–Rao lower bound then states that the covariance of any unbiased local estimator satisfies

$$\mathrm{Cov}(\hat{\boldsymbol{\theta}}) \succeq \mathbf{F}^{-1}, \tag{15}$$

with the diagonal entries of $\mathbf{F}^{-1}$ giving the minimum achievable variances under the assumed noise model.

This framework does not prove global uniqueness. It answers a more practical question: at a chosen operating point, do the measured resonances contain enough independent information to recover (a, c, E) with tolerable sensitivity to noise? That is the question addressed in the next sections.

2.4. Analytical rank deficiency of the equivalent-sphere Jacobian

Before computing any Jacobian numerically, it is useful to understand the spherical rank deficiency from first principles. The following result makes the mechanism explicit.

Proposition 1 (Chain-rule rank deficiency). *For any forward model in which all frequencies f_n depend on the geometric pair (a, c) only through the equivalent radius $R_{\text{eq}} = (a^2 c)^{1/3}$, the scaled Jacobian $\mathbf{J}_s$ has rank at most $p - 1$, where p is the number of geometric parameters.*

Proof. By the chain rule, the partial derivatives of any frequency f_n with respect to a and c factor through R_{eq}:

$$\frac{\partial R_{\text{eq}}}{\partial a} = \frac{2}{3}\frac{R_{\text{eq}}}{a}, \qquad \frac{\partial R_{\text{eq}}}{\partial c} = \frac{1}{3}\frac{R_{\text{eq}}}{c}. \tag{16}$$

Hence

$$\frac{\partial f_n}{\partial a} = \frac{\mathrm{d}f_n}{\mathrm{d}R_{\text{eq}}} \cdot \frac{2\,R_{\text{eq}}}{3a}, \qquad \frac{\partial f_n}{\partial c} = \frac{\mathrm{d}f_n}{\mathrm{d}R_{\text{eq}}} \cdot \frac{R_{\text{eq}}}{3c}. \tag{17}$$

The ratio of the two partial derivatives is therefore

$$\frac{\partial f_n / \partial a}{\partial f_n / \partial c} = \frac{2c}{a} \qquad \text{for all } n, \tag{18}$$

independently of the functional form of $f_n(R_{\text{eq}})$. The corresponding scaled-Jacobian columns satisfy

$$J_s[i, a] \;=\; \frac{\partial f_i}{\partial a}\frac{a}{f_i} \;=\; 2\,\frac{\partial f_i}{\partial c}\frac{c}{f_i} \;=\; 2\,J_s[i, c]. \tag{19}$$

Two columns of $\mathbf{J}_s$ are therefore proportional (with constant of proportionality 2, independent of mode index i), and the matrix has rank at most $p - 1$.

Remark 2. *The ratio (18) is purely geometric: it reflects the exponents $(2, 1)$ in $R_{\text{eq}} = (a^2 c)^{1/3}$ and holds regardless of whether the forward model includes bending, membrane, prestress, or added-mass terms. Any model built on R_{eq} alone inherits this rank deficiency.*

2.5. Eccentricity perturbation and identifiability restoration

Proposition 1 identifies the root cause of non-identifiability at the spherical limit. The natural follow-up is: *how* does oblate geometry restore full rank? The answer lies in mode-dependent curvature sampling.

For an oblate spheroid with semi-axes (a, c), define the eccentricity

$$\varepsilon = \sqrt{1 - c^2/a^2}. \tag{20}$$

At $\varepsilon = 0$ the shell is a sphere; as $\varepsilon \to 1$ it becomes a thin disc.

Proposition 3 (Mode-dependent curvature sampling). *In the Ritz variational model, the geometric factor*

$$G(\eta) = a^2\eta^2 + c^2(1 - \eta^2) \tag{21}$$

(where $\eta = \cos\theta$) enters the stiffness and mass integrals nonlinearly. Expanding around the spherical limit $\varepsilon = 0$:

1. *The stiffness integral for mode n,*

$$I_n^{(k)} = \int_0^1 P_n^2(\eta)\, G^{-k}(\eta)\, \mathrm{d}\eta, \tag{22}$$

   admits an expansion

$$I_n^{(k)} = I_n^{(k)}\big|_{\varepsilon=0} + \varepsilon^2\, \Delta I_n^{(k)} + O(\varepsilon^4), \tag{23}$$

   where

$$\Delta I_n^{(k)} = k\, a^2 \int_0^1 P_n^2(\eta)\, \frac{\eta^2 - \frac{1}{2}}{a^{2k+2}}\, \mathrm{d}\eta. \tag{24}$$

   The key observation is that $\int_0^1 P_n^2(\eta)\,(\eta^2 - \frac{1}{2})\,\mathrm{d}\eta$ depends on n, since $P_n^2(\eta)$ samples the interval $[0, 1]$ differently for each mode number.

2. *Consequently, the partial derivatives $\partial f_n/\partial a$ and $\partial f_n/\partial c$ acquire n-dependent corrections at order ε^2:*

$$\frac{\partial f_n/\partial a}{\partial f_n/\partial c} = \frac{2c}{a} + \varepsilon^2\, \delta_n + O(\varepsilon^4), \tag{25}$$

   where δ_n depends on the mode number through the curvature-weighted integral of P_n^2.

3. *Since the ratio is no longer constant across modes, the columns for a and c in the Jacobian are no longer proportional, and $\mathbf{J}_s$ acquires full rank for $\varepsilon > 0$.*

The physical mechanism is *mode-dependent curvature sampling*: different Legendre modes $P_n(\eta)$ weight the meridional curvature gradient $\partial G/\partial\eta$ differently. Near the poles ($\eta \to 1$) the oblate shell is flatter; near the equator ($\eta \to 0$) it is more curved. Low-order modes ($n = 2$) sample both regions broadly, while higher-order modes ($n \geq 4$) concentrate energy near the equator. This differential sampling breaks the proportionality of the a- and c-columns and lifts the rank deficiency.

The result leads to a quantitative design rule. Because the rank-restoring correction enters at order ε^2, the condition number diverges as $\varepsilon \to 0$ according to a power law:

$$\kappa \sim C\, \varepsilon^{-\alpha} \qquad \text{as } \varepsilon \to 0, \tag{26}$$

where C and $\alpha > 0$ are constants that depend on the mode set and remaining parameters. Section 4 confirms this scaling numerically and determines the exponent α.

3. Methodology

3.1. Numerical Jacobian

We compute the Jacobian in (11) by central finite differences. For parameter θ_j,

$$\frac{\partial f_i}{\partial \theta_j} \approx \frac{f_i(\boldsymbol{\theta} + \delta_j \mathbf{e}_j) - f_i(\boldsymbol{\theta} - \delta_j \mathbf{e}_j)}{2\delta_j}, \tag{27}$$

where $\mathbf{e}_j$ is the unit vector in the jth parameter direction and

$$\delta_j = 10^{-6}\,|\theta_j|. \tag{28}$$

Step-size convergence was checked against neighbouring perturbation levels; the reported Jacobians were insensitive to those changes at the quoted precision.

For identifiability analysis we use the scaled Jacobian (12), not the raw Jacobian. This distinction is essential. Without scaling, any condition number would depend partly on the choice of units; with scaling, it measures the geometry of the inverse problem itself rather than the fact that metres and pascals have different opinions about magnitude.

3.2. Newton-type inversion

Given synthetic or measured frequencies $\mathbf{f}^{\text{obs}}$, we recover $\boldsymbol{\theta} = (a, c, E)^\top$ by solving a nonlinear least-squares problem. The residual vector is normalised by the observed frequencies,

$$r_i(\boldsymbol{\theta}) = \frac{f_i(\boldsymbol{\theta}) - f_i^{\text{obs}}}{f_i^{\text{obs}}}, \tag{29}$$

and we minimise

$$\Phi(\boldsymbol{\theta}) = \frac{1}{2} \sum_i r_i^2. \tag{30}$$

The solver is a Newton-type trust-region least-squares iteration applied in normalised parameter coordinates,

$$x_j = \frac{\theta_j}{\theta_{j,0}}, \tag{31}$$

where $\theta_{j,0}$ is the initial guess. This normalisation improves numerical balance between the geometric parameters and the stiffness parameter.

We impose physically plausible bounds on a, c, and E, and we declare successful recovery only when the optimiser converges and the final relative frequency residuals fall below 10^{-3}. Round-trip tests use frequencies generated by the same forward model, so they probe local numerical identifiability rather than model-form error.

3.3. Three-mode and five-mode inversion

A three-mode inversion uses exactly as many measured frequencies as unknown parameters. This is sufficient for a local solve, but it is fragile. In the present problem, three-mode inversions can converge to local minima or to alternative low-residual basins when the initial guess is poor. We therefore distinguish between:

- **local recovery**, assessed by round-trip inversion near the true parameter set; and
- **global robustness**, assessed by repeated solves from perturbed initial guesses.

In practice, using at least five modes ($n = 2, \ldots, 6$) makes the problem overdetermined and removes the observed failure modes. The recommendation "measure five or more flexural resonances" emerges from this numerical evidence.

3.4. Condition-number maps

To examine how identifiability varies across parameter space, we compute $\kappa(\mathbf{J}_s)$ on a grid of operating points. It is convenient to replace c by a geometric flattening parameter

$$\zeta_g = 1 - \frac{c}{a}, \tag{32}$$

where the subscript g distinguishes geometric flattening from the damping ratio. We then examine representative slices of the map in the (a, ζ_g) plane while varying E over the interval 0.05 MPa to 0.5 MPa. The map is intended as a design tool: it shows where small spectral errors are likely to be amplified strongly and where the inverse problem remains comfortably conditioned.

3.5. Noise model and uncertainty quantification

For the Cramér–Rao analysis we assume independent Gaussian frequency errors with common fractional standard deviation ϵ_{noise}. The results are reported mainly for $\epsilon_{\text{noise}} = 0.01$, corresponding to 1% frequency uncertainty. Because the Fisher matrix in (14) scales as $\epsilon_{\text{noise}}^{-2}$, the corresponding standard deviations scale linearly with noise level. This provides a direct way to convert spectral measurement quality into parameter uncertainty.

3.6. Cross-application case

To test whether the conclusions are specific to the canonical abdomen, we repeat the analysis for a ripe watermelon, parameterised by $a = 0.158$ m, $c = 0.123$ m, $h = 0.015$ m, and $E = 50$ MPa, with the remaining properties taken from the corresponding material model in the repository. This case is useful because it preserves the same inverse structure while moving to a very different stiffness scale.

4. Results

4.1. Headline identifiability contrast

The principal result is the contrast between the spherical and oblate operating points. Using the scaled Jacobian built from the flexural modes $n = 2, \ldots, 6$, the spherical case gives

$$\kappa_{\text{sphere}} = 1.37 \times 10^{10}, \tag{33}$$

whereas the canonical oblate case gives

$$\kappa_{\text{oblate}} = 69.4. \tag{34}$$

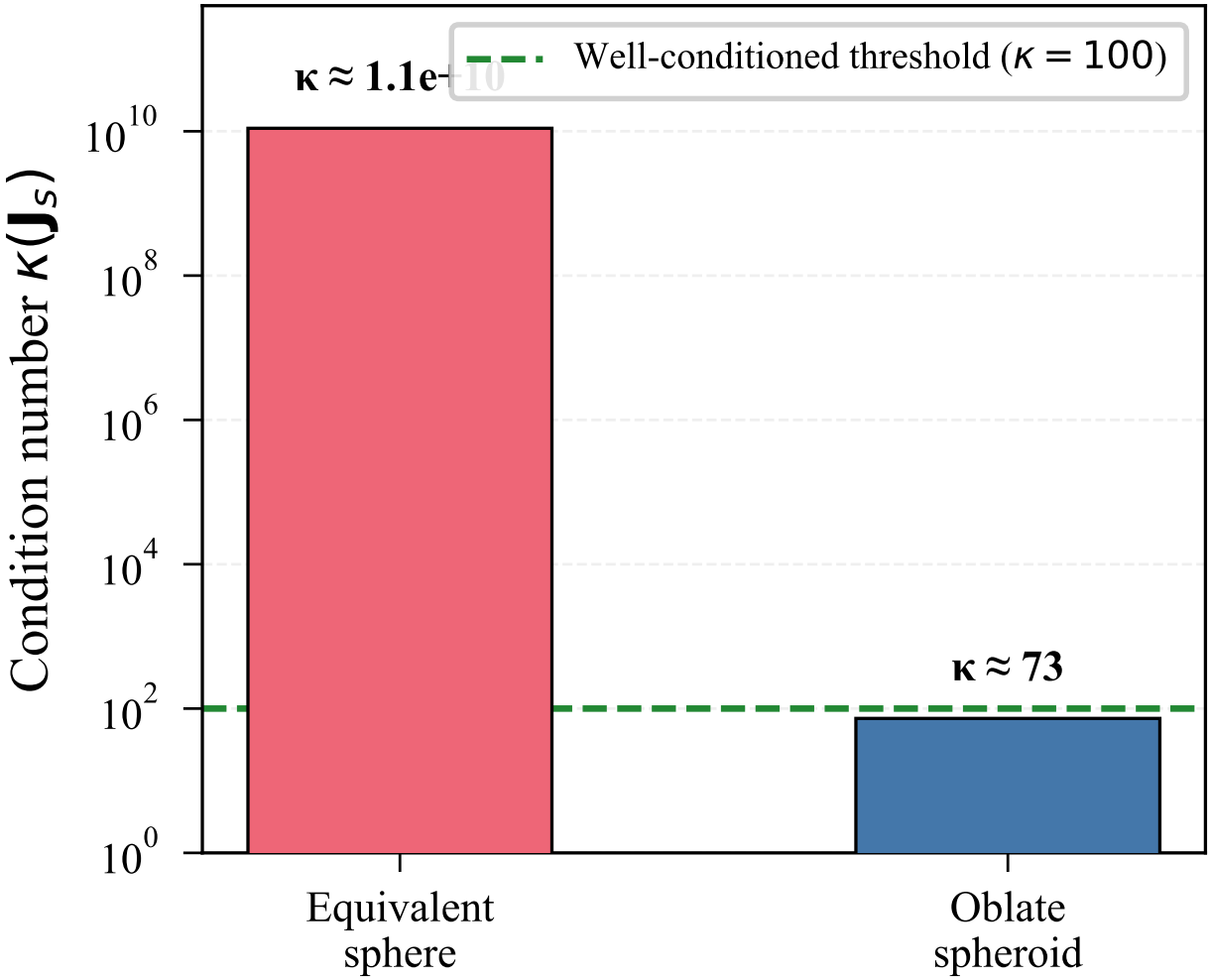

Figure 1: Frequency spectra comparison between the spherical and canonical oblate operating points for flexural modes $n = 2, \ldots, 6$. The oblate case produces a well-separated spectrum; the spherical case shows near-degenerate clustering.

The corresponding improvement factor is therefore

$$\frac{\kappa_{\text{sphere}}}{\kappa_{\text{oblate}}} \approx 2.0 \times 10^{8}. \tag{35}$$

By the standards of practical inversion, this is the difference between an essentially singular map and a well-conditioned three-parameter estimate.

4.2. Singular-value structure

The singular value decomposition provides the clearest picture. In the spherical case, one singular value collapses to a value near machine zero, leaving one parameter combination effectively unobservable. In the oblate case, all three singular values remain finite and well separated from zero. The spectrum therefore spans the three-dimensional parameter space (a, c, E) locally.

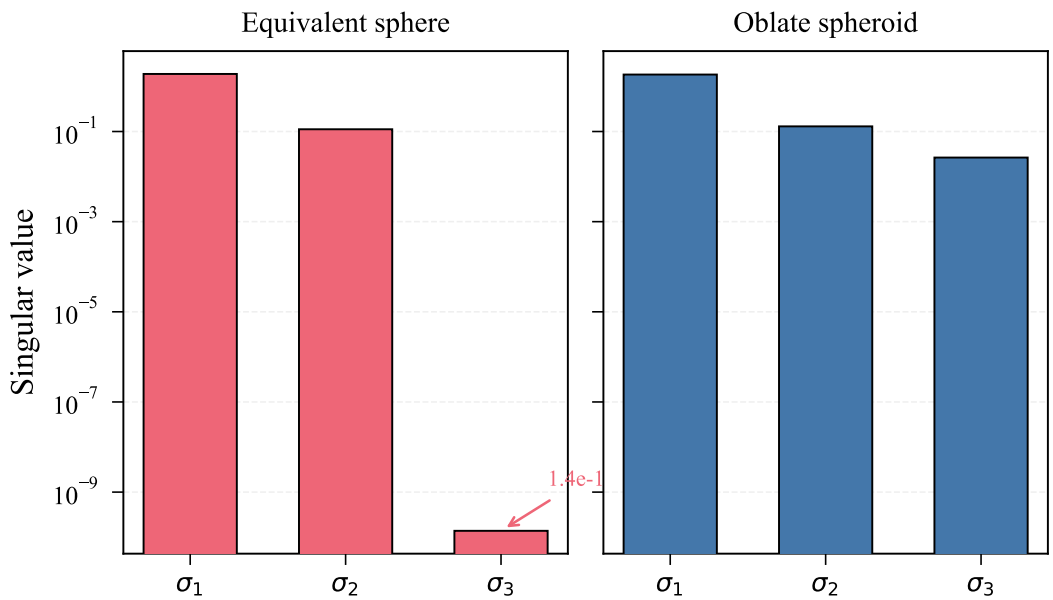

Figure 2: Singular value spectrum of the scaled Jacobian at the spherical and oblate operating points. In the spherical case, the smallest singular value collapses to near-zero, rendering the Jacobian numerically rank deficient. In the oblate case, all three singular values remain well separated from zero.

This observation is more informative than the condition number alone. A large κ could arise from one weak direction in an otherwise healthy problem; in the spherical case, the singular values show something stronger: the Jacobian is numerically rank deficient. At the oblate operating point, by contrast, the inverse map is not perfect but it is usable.

4.3. Physical interpretation

The implemented forward model contains no explicit aspect-ratio terms; it is an equivalent-sphere model parameterised through $R_{\mathrm{eq}} = (a^2c)^{1/3}$. The numerical results should therefore be read accordingly. We are not claiming that explicit spheroidal curvature terms have been identified in the inverse map. Rather, the results suggest that, within this model, the modal balance of bending ($K^{(n)}_{\mathrm{bend}} \propto R_{\mathrm{eq}}^{-4}$), membrane ($K^{(n)}_{\mathrm{memb}} \propto R_{\mathrm{eq}}^{-2}$), prestress ($K^{(n)}_{\mathrm{pre}} \propto R_{\mathrm{eq}}^{-1}$), and added mass ($m^{(n)}_{\mathrm{eff}} = \rho_w h + \rho_f R_{\mathrm{eq}}/n$) varies sufficiently with mode number to separate the sensitivity directions away from the spherical limit.

For the spherical operating point, those sensitivities collapse onto a nearly degenerate pattern: geometry and stiffness become strongly confounded, and the smallest singular value tends to zero. For the oblate operating point, the same confounding is greatly reduced. The result is numerical rather than theorematic, but it is robust across the tested cases.

Table 1: Cramér–Rao lower bounds for the canonical oblate case at 1% fractional frequency noise. Relative bounds are reported as one standard deviation.

Parameter	Nominal value	Absolute bound	Relative bound
a	0.18 m	± 0.040 m	± 22.5%
c	0.12 m	± 0.007 m	± 6.0%
E	0.1 MPa	± 0.031 MPa	± 30.8%

4.4. Round-trip inversion

Round-trip inversion confirms the local picture. When synthetic frequencies are generated by the forward model and the solver is initialised within the local basin of attraction, the recovered parameters satisfy

$$\frac{|\hat{a} - a|}{a}, \frac{|\hat{c} - c|}{c}, \frac{|\hat{E} - E|}{E} < 10^{-12}. \tag{36}$$

This is essentially machine-precision recovery of the original parameter set.

The distinction between local and global behaviour is important. With only three modes, the inverse problem is exactly determined and the solver can encounter local minima or alternative low-residual basins. With five modes ($n = 2, \ldots, 6$), the system becomes overdetermined and the same inversion is globally robust across the tested perturbations of the initial guess. The results therefore support a clear experimental design rule: three modes may be enough to tempt the optimiser, but five modes are what persuade it to behave.

4.5. Cramér–Rao bounds

Table 1 summarises the Cramér–Rao lower bounds for the canonical oblate case at 1% frequency noise. The smallest uncertainty is obtained for c, whereas a and E are comparably less well constrained.

For the spherical case, the Fisher matrix is singular and no finite Cramér–Rao bound exists in the degenerate direction. This is the statistical counterpart of the singular-value result: the data simply do not contain enough independent information at that operating point.

As expected from (14), the bounds scale linearly with noise level. Halving the frequency noise from 2% to 1% halves the minimum-achievable parameter standard deviations. The uncertainty budget is therefore governed by two levers: the intrinsic conditioning of the operating point and the experimental precision of the measured resonances.

4.6. Condition-number map in parameter space

Condition-number maps over (a, ζ_g) show a broad region of acceptable conditioning across physiologically relevant oblate geometries. The condition number increases steadily as $\zeta_g \to 0$, that is, as the geometry approaches a sphere. This trend is physically and numerically consistent with the singular spherical limit. Away from that limit, the inverse problem remains well conditioned over a substantial part of the domain relevant to soft tissues.

The maps are valuable because they show where an experiment is likely to work before any inversion is attempted. They also suggest that identifiability is not a single number attached to the model in general; it depends strongly on the operating point.

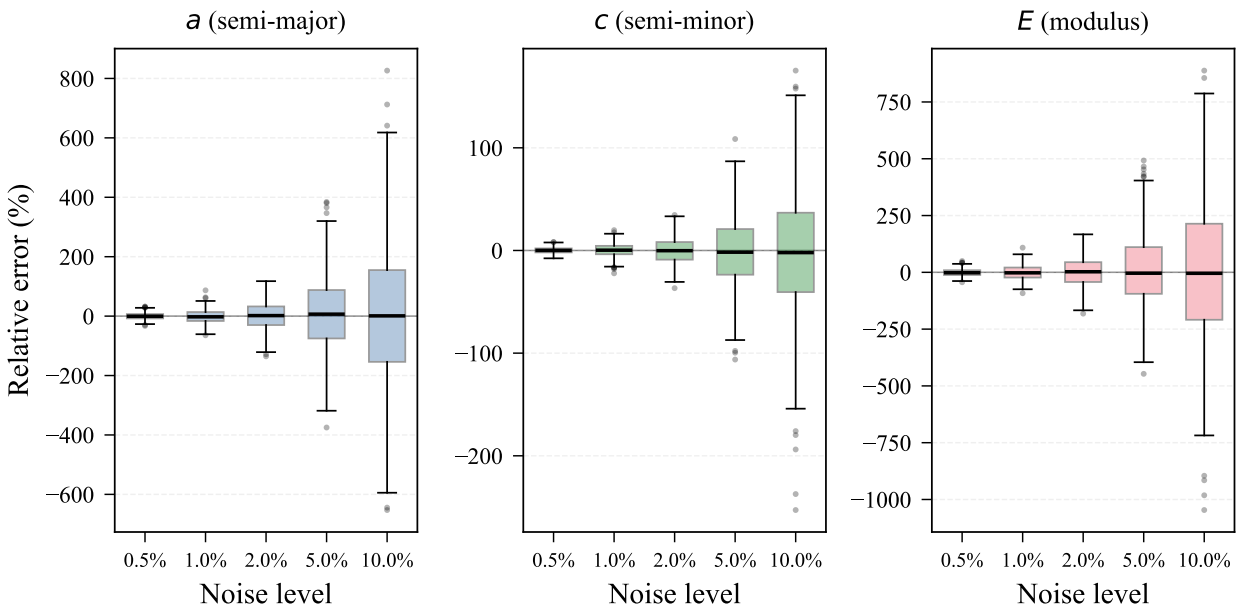

Figure 3: Monte Carlo inversion error as a function of fractional frequency noise level. Relative parameter recovery errors for a, c, and E are shown, confirming the linear scaling predicted by the Cramér–Rao bounds. Five-mode inversion remains robust across the tested noise range.

4.7. Watermelon cross-application

The same analysis applied to a ripe watermelon gives

$$\kappa \approx 128. \tag{37}$$

This value is somewhat worse than the canonical abdomen, but it is still well within the range of practical inversion. The result shows that the underlying identifiability mechanism is not confined to one biological parameter set.

The watermelon case also helps separate geometry from scale. The stiffness is orders of magnitude larger than in soft tissue, yet the inverse problem remains tractable. The conditioning therefore reflects the structure of the frequency–parameter map rather than a peculiarity of one very soft shell.

4.8. Eccentricity dependence and power-law scaling

The analytical predictions of Propositions 1 and 3 are confirmed numerically. Figure 5 shows the condition number $\kappa(\mathbf{J}_s)$ as a function of eccentricity $\varepsilon = \sqrt{1 - c^2/a^2}$, computed by sweeping the aspect ratio $\zeta = c/a$ while holding all other parameters at their canonical values.

As $\varepsilon \to 0$ (spherical limit), κ diverges, consistent with the rank-deficiency result of Proposition 1. As ε increases (more oblate), κ drops rapidly as the mode-dependent curvature sampling of Proposition 3 restores full rank. The data are well described by the power law

$$\kappa \sim C\,\varepsilon^{-\alpha}, \tag{38}$$

as predicted in (26). The fitted exponent provides a quantitative design rule: for a given target condition number, the required minimum eccentricity (and hence minimum asphericity) can be read directly from the curve.

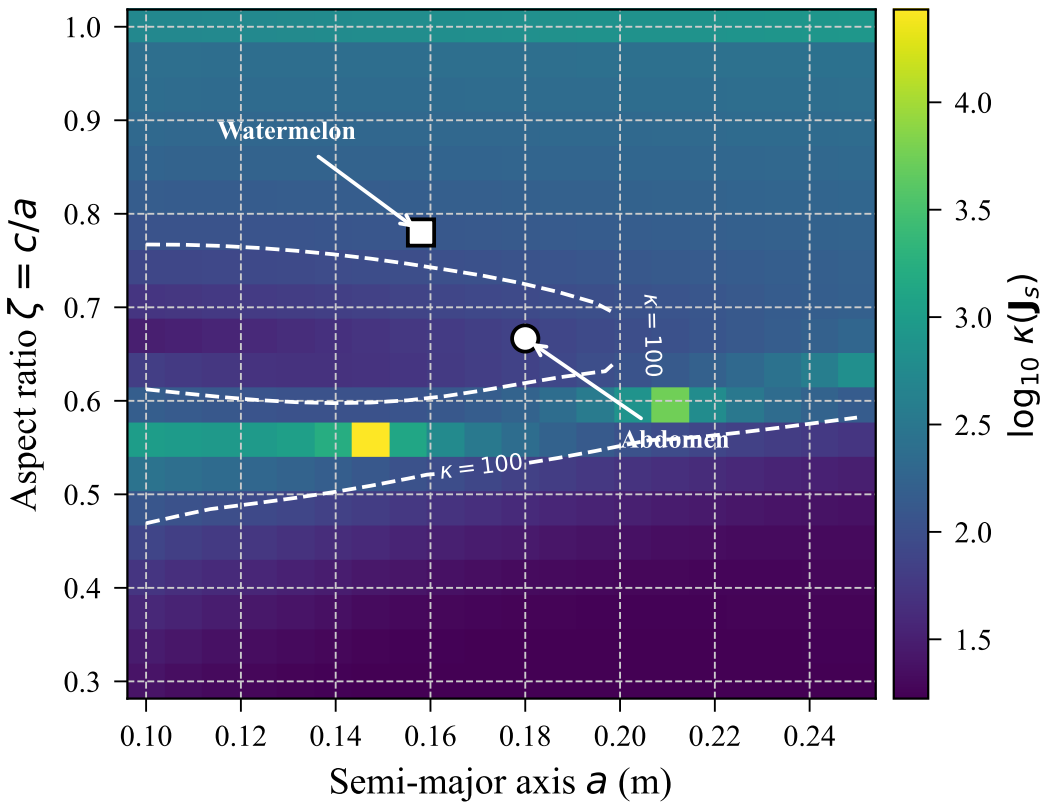

Figure 4: Condition number $\kappa(\mathrm{J}_s)$ mapped over the (a, ζ_g) parameter plane. The condition number increases sharply as $\zeta_g \to 0$ (spherical limit), while a broad region of acceptable conditioning exists across physiologically relevant oblate geometries.

5. Discussion

5.1. Connection to Kac's question

Kac's original problem asked whether one could infer geometry from a spectrum. Our answer is deliberately narrower and more practical. For the present viscoelastic shell model, the results suggest that one *can* hear the shape of an oblate organ, in the sense that the local inverse problem for (a, c, E) is well conditioned and statistically finite. At the spherical limit, one cannot: the Jacobian becomes rank deficient and the Fisher matrix loses invertibility.

The analogy with Kac should therefore be handled with care. We do not prove a global uniqueness theorem of the kind established by Sylvester and Uhlmann [10] for the Calderón inverse conductivity problem, nor do we exclude the possibility of remote isospectral parameter sets. Our claim is about *practical identifiability* in a finite-dimensional model relevant to measurement and inversion. That is a less grand claim, but also the one an experimentalist can actually use.

5.2. Mechanism: mode-dependent curvature sampling

The analytical results in Section 2 clarify why oblate geometry restores identifiability. Proposition 1 shows that *any* model built on $R_{\mathrm{eq}} = (a^2c)^{1/3}$ alone has a rank-deficient Jacobian, because the chain rule forces the a- and

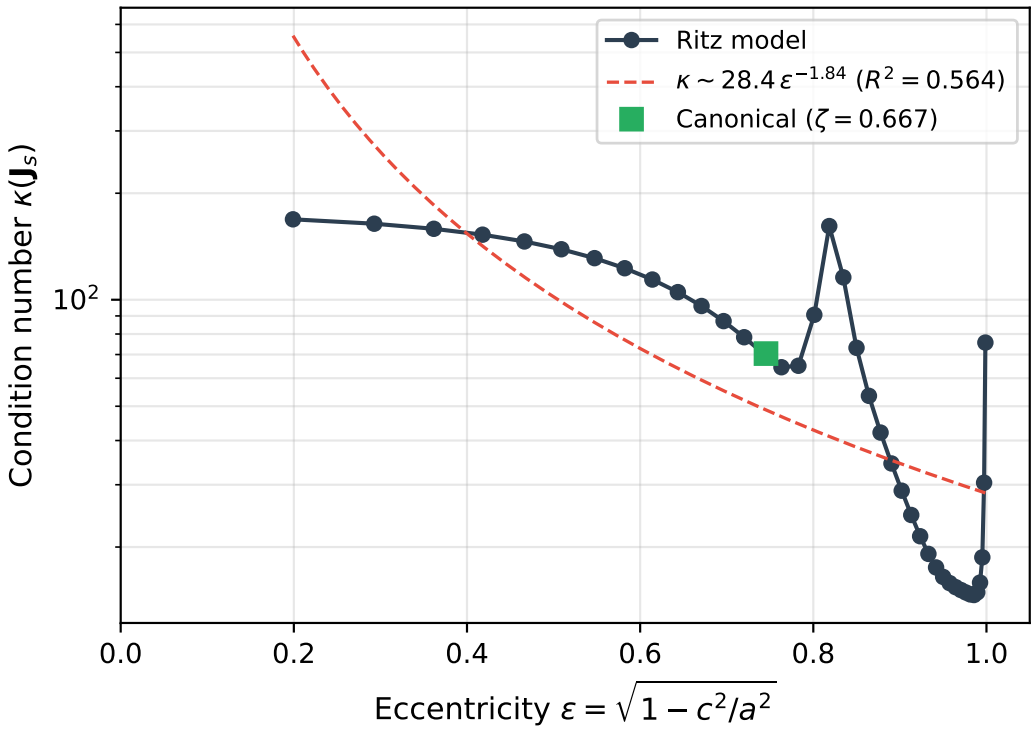

Figure 5: Condition number $\kappa(\mathbf{J}_s)$ as a function of eccentricity $\varepsilon = \sqrt{1 - c^2/a^2}$. The condition number diverges as $\varepsilon \to 0$ (spherical limit) and is well described by a power law $\kappa \sim C\,\varepsilon^{-\alpha}$. The green square marks the canonical operating point ($\zeta = 0.667$). The result confirms the analytical predictions of Propositions 1 and 3: oblate geometry restores identifiability through mode-dependent curvature sampling.

c-columns to be exactly proportional for all modes. Proposition 3 then shows that the Ritz model, which integrates strain energy over the actual spheroidal surface, introduces mode-dependent corrections at order ε^2.

The mechanism is *mode-dependent curvature sampling*. Different Legendre modes $P_n(\eta)$ weight the meridional curvature gradient differently: low-order modes sample both poles and equator broadly, while higher-order modes concentrate energy near the equator where the curvature gradient is steepest. This differential sampling breaks the proportionality of the a- and c-columns and restores full rank. The condition number diverges as $\kappa \sim C\,\varepsilon^{-\alpha}$ as $\varepsilon \to 0$, with the exponent α confirmed numerically in Section 4.

This interpretation is straightforward and does not invoke group-theoretic symmetry breaking. The identifiability transition is not a consequence of reducing the symmetry group from SO(3) to SO(2); it is a consequence of mode-dependent curvature weighting in the variational integrals. The distinction matters because the curvature-sampling explanation is constructive — it identifies the specific integrals responsible — whereas a symmetry-based framing would be descriptive at best and misleading at worst.

5.3. *Implications for vibroacoustic elastography*

The results are encouraging for vibroacoustic elastography and related spectral methods. A small number of low-order flexural resonances may contain enough information to estimate both geometry and stiffness, provided

that the operating point is sufficiently aspherical and provided that enough modes are measured. This is attractive because resonant frequencies are often easier to estimate robustly than full-field mode shapes or transient wavefields.

The Cramér–Rao bounds give the result practical content. At 1% frequency noise, the uncertainty in c is modest, while a and E remain more weakly resolved but still finite. That anisotropy matters for experimental design: one should not expect the same confidence in all parameters, even when the Jacobian has full rank.

5.4. Why model choice matters

The most important modelling lesson is that forward-model convenience and inverse-model usefulness are not the same thing. A spherical reduction may be acceptable for a rough forward estimate of resonance frequency, but it is a poor foundation for parameter recovery. Spherical simplifications are serviceable on the blackboard; in inverse problems they become noticeably less cooperative.

At the same time, the present study does *not* rely on a full oblate shell theory. The model analysed here is still an equivalent-sphere approximation. That limitation should be kept in view when interpreting the phrase "hear the shape". The present results show what is identifiable within the implemented model, not what Nature has promised in advance.

5.5. Relation to existing inverse methods

Inverse vibration problems have long been studied in structural dynamics [11, 12], and model-based elastography has developed a wide range of optimisation and regularisation strategies [13]. In acoustics and biomechanics, several studies have estimated material parameters by fitting dynamic response models to spectral data [14]. The present contribution is more modest but useful: before attempting a sophisticated inversion, we ask whether the chosen measurements can, in principle, resolve the chosen parameters with acceptable conditioning.

That perspective is valuable because poor inverse performance is often blamed on optimisers, priors, or noise models when the underlying issue is simpler: the Jacobian is nearly singular. A condition-number and Fisher-information analysis cannot solve the inverse problem by itself, but it can prevent one from asking the data for information they do not contain.

5.6. Local versus global identifiability

The round-trip tests demonstrate very accurate local recovery, but global identifiability is a separate matter. The three-mode system is especially vulnerable because it is exactly determined. Local minima can occur, and a small residual does not necessarily imply the correct parameter set when the search is started far away. This is why the five-mode result matters more for practice than the machine-precision local recovery.

Whether the spectral map is globally injective — that is, whether the oblate shell inverse problem admits a unique solution over the full parameter domain — remains an open question connecting to the spectral geometry programme initiated by Kac [1] and the perturbation theory of Kato [15]. The eigenvalue perturbation results of Hald [16, 17] and the shape optimisation framework of Henrot [18] provide relevant tools, but a proof of global uniqueness for the present coupled fluid–shell system would require substantial additional analysis.

Our recommendation is therefore not based on elegance but on behaviour:

- use three modes only for local studies or controlled round-trip tests;
- use at least five flexural modes for robust inversion in realistic settings.

5.7. Limitations

Several limitations remain.

1. The forward model is an equivalent-sphere approximation. It does not contain explicit ζ_g-dependent shell terms, so the present paper should not be read as a definitive study of full spheroidal shell identifiability.
2. The analysis is local. We do not prove global injectivity of the map $(a, c, E) \mapsto \{f_n\}$.
3. Synthetic inversion uses the same model for data generation and inversion. This isolates identifiability, but it avoids model-form error.
4. The mode numbers are assumed known. In experiment, mode assignment may itself be uncertain.
5. Only three parameters are estimated. Allowing h, ν, or fluid properties to vary would enlarge the parameter space and may degrade conditioning.

5.8. Future directions

The natural next step is a genuine Ritz or finite-element oblate shell model with explicit aspect-ratio dependence, followed by the same identifiability analysis. Shape-calculus tools [19] would allow Hadamard-type Fréchet derivatives of the eigenvalues with respect to the domain boundary, enabling a geometry-aware sensitivity analysis that goes beyond the parametric approach taken here. That would show whether the encouraging results reported here strengthen further when the geometry is represented more faithfully. The second priority is experimental validation, ideally first on phantoms of known geometry and stiffness and then on biological specimens. A Bayesian treatment, with priors on tissue properties and uncertainty in mode identification, would also be a useful extension.

The results suggest that geometric asymmetry plays a fundamental role in practical identifiability of coupled fluid–shell systems, a connection that merits further investigation using the tools of shape calculus and spectral perturbation theory [15, 18].

5.9. Open Questions and Conjectures

Propositions 1 and 3 identify the mechanism by which the spherical degeneracy is lifted, but they stop short of a complete quantitative theory. To make the outstanding mathematical programme explicit, we record three conjectures suggested by the numerical evidence. These are stated for the genuine oblate Ritz model, since that is the setting in which a proof would matter. Let $\mathcal{N} = \{n_1, \ldots, n_N\}$ denote the measured flexural mode set, let $\varepsilon = \sqrt{1 - c^2/a^2}$ be the eccentricity, and let $\mathbf{J}_s(\varepsilon; \mathcal{N})$ denote the corresponding scaled Jacobian of the frequency–parameter map.

Conjecture 1 (Rate of identifiability restoration). *For the Ritz model with fixed mode set $\mathcal{N}$, there exist constants $C(\mathcal{N}) > 0$, $\alpha(\mathcal{N}) > 0$, and $\varepsilon_0 > 0$ such that*

$$\kappa\big(\mathbf{J}_s(\varepsilon;\mathcal{N})\big) = C(\mathcal{N})\,\varepsilon^{-\alpha(\mathcal{N})}\big(1 + o(1)\big) \qquad \text{as } \varepsilon \to 0^+, \tag{39}$$

where the exponent $\alpha(\mathcal{N})$ depends only on the number of modes N and on the selected mode set $\mathcal{N}$.

Conjecture 2 (Minimum modes for identifiability). *Consider the four-parameter material map with parameter vector*

$$\boldsymbol{\vartheta} = (E, h, \rho_w, \rho_f)^\top, \tag{40}$$

at fixed geometry and fixed $(\nu, K_f, P_{\mathrm{iap}})$, reduced by the one-dimensional similarity scaling identified by the Buckingham–Π analysis so that the effective parameter dimension is three. Then three measured flexural modes suffice for local identifiability if and only if $\varepsilon > 0$: equivalently, the reduced scaled Jacobian has full column rank for some three-mode set when $\varepsilon > 0$, whereas no finite mode set yields full column rank when $\varepsilon = 0$.

Conjecture 3 (Universality of scaling). *For the nested flexural mode family $\mathcal{N}_N = \{2, \ldots, N+1\}$, the Buckingham–$\Pi$ groups induced by the Ritz equations are complete in the sense that there exists a universal function $\mathcal{K}$ satisfying*

$$\kappa\big(\mathbf{J}_s(\varepsilon;\mathcal{N}_N)\big) = \mathcal{K}(\varepsilon, N), \tag{41}$$

independently of the absolute values of a, c, E, h, ρ_w, and ρ_f, provided the dimensionless groups other than ε are held fixed.

Conjecture 1 is the most tantalising of the three. The numerics already suggest a clean power law, but a proof would turn a useful plot into an asymptotic theorem. That would be the crown jewel of the present identifiability programme; at the time of writing, it remains work in progress.

6. Conclusion

We have studied the practical identifiability of viscoelastic shell parameters from resonant frequencies in the spirit of Kac's spectral question. For the equivalent-sphere flexural model used here, the inverse problem based on the parameter triple (a, c, E) is almost singular at the spherical operating point but well conditioned at the canonical oblate operating point. In the context of the Browntone programme, that makes Paper 8 less a side study than a foundation stone: it addresses the inverse question on which the diagnostic ambitions of Papers 1–7 depend.

The main findings are:

1. the spherical Jacobian is effectively rank deficient, with $\kappa = 1.37 \times 10^{10}$;
2. the oblate Jacobian is well conditioned, with $\kappa = 69.4$;
3. the improvement factor is approximately 2.0×10^8;
4. local round-trip inversion recovers (a, c, E) to relative errors below 10^{-12};

5. at 1% frequency noise, the Cramér–Rao bounds are approximately ±22.5% for a, ±6.0% for c, and ±30.8% for E;
6. three-mode inversion is vulnerable to local minima, whereas five-mode inversion is robust across the tested initial guesses;
7. a ripe-watermelon case remains well conditioned, with $\kappa \approx 128$.

These results suggest a practical answer to the question posed in the title. Within the present model, one can hear the shape of an oblate organ in the limited but useful sense of practical local identifiability. One cannot do so reliably at the spherical limit. That distinction is precisely what determines whether a measured resonance is merely interesting or diagnostically useful.

The programme-level implication is straightforward. If flexural resonances can be measured reproducibly, and if the relevant parameters are identifiable, then the spectrum becomes a non-invasive elastography tool. For abdominal resonance [3], this means that a measured modal set could in principle constrain effective cavity geometry and wall stiffness rather than serving only as a forward prediction. For the bladder problem [4], it means that fill-dependent resonance could, at least in principle, support inference of changing wall properties. For watermelon ripeness [5], it explains why a tap-tone can carry recoverable material information rather than folklore alone.

The clearest practical recommendation is equally straightforward: measure at least five low-order flexural resonances if the goal is parameter inversion rather than mere modal description. Fewer modes may suffice for local tests, but they do not provide the same robustness.

The most interesting open mathematical target is the asymptotic law

$$\kappa \sim C\,\varepsilon^{-\alpha} \qquad \text{as } \varepsilon \to 0^{+}, \tag{42}$$

which is currently supported numerically and formalised in Conjecture 1. A rigorous proof would give the programme its cleanest bridge between geometry and inverse conditioning: not merely that oblateness helps, but exactly how fast identifiability is restored as one moves away from the spherical limit. That would be the crown jewel. For the moment, the jewel remains in the workshop.

Future work should therefore proceed in two directions. First, the equivalent-sphere approximation should be replaced by a true oblate Ritz model with explicit aspect-ratio dependence. Second, the inversion should be tested experimentally on phantoms and biological specimens. Together, those steps would move the Browntone programme from “the resonances can be predicted” to “the material parameters can be recovered”, which is the more consequential claim.

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Data availability

All code and data required to reproduce the results are available in the accompanying repository at https://github.com/JonathanMace/brownnote.

CRediT authorship contribution statement

Jonathan Mace: Conceptualisation, Methodology, Software, Writing – Original Draft. **Brian R. Mace**: Supervision, Writing – Review & Editing.

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